

Lab 3: SQL for Finance Data

Instructions

1. What This Lab is About

This lab transitions you from basic spreadsheet manipulation to professional grade financial data management using SQL. You will learn how to interact with financial databases to audit records, reconcile balances, and extract actionable insights from raw data, bridging the gap between technical data manipulation and core finance functions.

Before you begin: Follow the setup instructions in the **README** file included in the lab package. Once your notebook is running, return to this document to understand the learning objectives and expected outcomes.

2. Learning Outcomes

- C2: SQL
- D2: Data preparation and cleaning

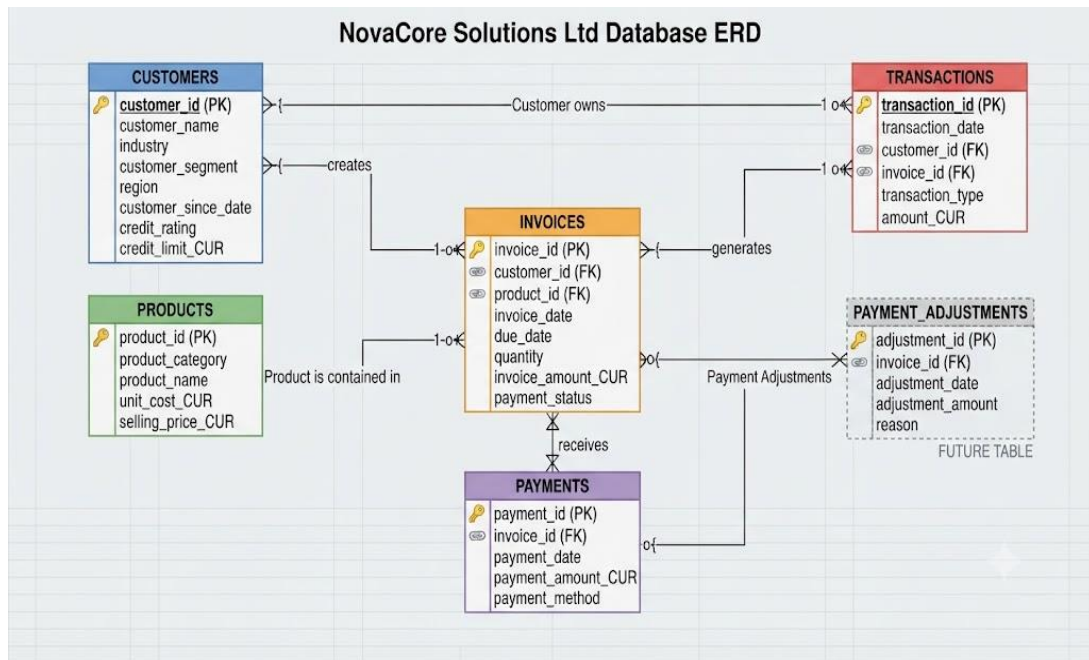
3. Study material covered

- Chapter 5: Programming for Business
- Chapter 6: Data Modelling Process and Techniques

4. Expected results

By the end of this lab, you should have:

- A summary of the NovaCore database tables and record counts.
- A payment_adjustments table created and populated using SQL.
- Data quality outputs showing missing values, duplicate checks and standardised customer segments.
- SQL join outputs identifying invoices without matching payment records.
- Finance analysis outputs including an invoice KPI summary, an outstanding balance report and a top 10 high-value invoices review.



5. Employability tips

- **SQL as an Audit Essential:** SQL is a non-negotiable skill for modern audit and assurance; understanding how to query databases directly replaces slow, error prone manual spreadsheet cross-referencing.
- **Data Cleaning for FP&A:** Financial Planning & Analysis (FP&A) roles rely on clean data. Mastering `COALESCE` and standardisation functions makes you faster and more accurate than peers relying on manual 'Find & Replace'.
- **Building Institutional Knowledge:** Being able to design and populate your own tables (as done in this lab) is a key skill for financial modelling where standard system outputs are insufficient.

If you experience setup, database connection or file access issues, refer to the Troubleshooting Guide in the README file included in the lab package.